Prodigy Anesthesia Quick Review

Combined Spinal/Epidural

What is the traditional method for performing a combined spinal/epidural?	An epidural needle is placed in the epidural space. A spinal needle is then passed through the epidural needle and into the subarachnoid space. The spinal anesthetic is administered and the spinal needle is withdrawn completely. The epidural catheter is then threaded through the epidural needle and can then be used to increase the duration of the neuraxial block intraoperatively or used for postoperative analgesia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 915.*
What is the advantage to performing a combined spinal/epidural?	It provides the rapid onset and dense block of a spinal anesthetic with the duration and flexible dosing provided by an epidural catheter. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 915.*
How does the selection of the spinal needle for a combined spinal/epidural affect the incidence of postdural puncture headache?	The use of 24 to 27 gauge pencil-point needles (Whitacre, Sprotte, or Gertie Marx) helps reduce the incidence of postdural puncture headache. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1085.*
What is the appropriate spinal needle length for performing a one-level, combined spinal/epidural anesthetic?	One of the factors associated with a decreased success rate on the spinal portion of the procedure is a spinal needle that is too short. The spinal needle should be 7-15 millimeters longer than the epidural needle. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1096.*
How does the use of intrathecal opioids in a combined spinal/epidural affect the incidence of postdural puncture headache?	The use of opioids has been shown to reduce the incidence of postdural puncture headache in combined spinal/epidural anesthesia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097.*
How does patient position affect the odds for success when performing a combined spinal/epidural anesthetic?	The odds for success are higher when the procedure is performed with the patient in a sitting, rather than lateral position. This is probably due to the higher hydrostatic pressure in the subarachnoid space in the sitting position, which makes the detection of CSF more likely. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097.*

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What factors are associated with an increased risk that the epidural catheter could migrate into the subarachnoid space after a combined spinal/epidural?	Dural puncture with a 25-gauge needle does not increase the risk, but puncture of the dura with an 18-gauge Tuohy needle increases the risk that the epidural catheter could migrate into the subarachnoid space. Other factors associated with an increased risk include patient movement, undetected dural puncture, and diffusion of the epidural dose into the subarachnoid space. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1096-1097.*
When performing the epidural portion of a combined spinal/epidural, should you use air or saline to determine entry into the epidural space and why?	The epidural space should be identified with the use of air because the use of saline may create a potential confusion between saline and CSF. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 854.*
How does injection of local anesthetic into the epidural catheter affect the spread of the spinal anesthetic and why?	Injection of solution through the epidural catheter is known to increase the spread of the spinal anesthetic. It is believed that the increased volume injected into the epidural space compresses the subarachnoid space, resulting in an increase in pressure that 'pushes' the local anesthetic in the subarachnoid space higher. Another cause is the leakage of local anesthetic from the epidural space into the subarachnoid space via the dural puncture made for the spinal anesthetic. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1097.*
If you are performing an epidural anesthetic and inadvertantly puncture the dura, can you convert the epidural to a combination spinal/epidural?	Yes, this a common technique for dealing with inadvertant dural puncture. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 298.*
How is a two-level combined spinal/epidural performed?	A two-level combined spinal/epidural is performed by first placing the epidural catheter, then performing a spinal anesthetic one or two interspaces below the epidural. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1095.*
What is the advantage of performing a two-level combined spinal/epidural?	It allows the anesthesia provider the ability to test the function of the epidural catheter prior to placing the spinal anesthetic. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1095.*
What are the disadvantages of performing a two-level combined spinal/epidural?	It can be difficult to distinguish between the epidural test dose and CSF when performing the spinal anesthetic. CSF flow may be hindered by the compression of the dural sac by the epidural test dose. The epidural catheter can be cut by the spinal needle. There is a risk of dural puncture with the epidural catheter. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1095.*

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What is a disadvantage of performing a continuous spinal/epidural anesthetic?

It is asssociated with a high incidence of postdural puncture headache.

Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 298.

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